

Product Requirements Document (PRD)

HackGuard AI

Secure AI-Powered Hackathon Hosting & Automated Evaluation Engine

Version: 1.0

Project Type: AI + Full Stack + DevOps + Cybersecurity

Domain: General

Target Users: Universities, Enterprises, Developer Communities, Hackathon Organizers

1. Product Overview

HackGuard AI is an AI-powered hackathon management platform designed to automate the complete lifecycle of a hackathon—from participant registration to project evaluation and winner announcement.

The platform combines secure code execution, AI-powered project analysis, plagiarism detection, automated scoring, and real-time leaderboards into one unified system.

Its primary objective is to reduce manual judging effort, eliminate submission fraud, and provide transparent, scalable, and fair project evaluation.

2. Problem Statement

Current hackathon platforms face several challenges:

- Manual evaluation consumes significant time and is prone to bias.
 - Organizers use multiple disconnected tools for registration, submissions, communication, and judging.
 - Code plagiarism and copied GitHub repositories are difficult to detect.
 - AI-generated projects are often submitted without genuine implementation effort.
 - Lack of secure environments to execute participant code.
 - Delayed result announcements due to manual review.
-

3. Vision

To create an intelligent, secure, and scalable hackathon platform capable of evaluating projects automatically while maintaining fairness, transparency, and security.

4. Objectives

- Automate project evaluation.
 - Detect plagiarism across submissions.
 - Analyze project documentation using AI.
 - Execute submitted projects securely.
 - Generate transparent scores.
 - Reduce judging time by more than 90%.
 - Provide real-time rankings.
 - Simplify hackathon management.
-

5. Target Users

Participants

- Students
- Developers
- Startup Teams
- Professionals

Organizers

- Universities
- Colleges
- Tech Communities
- Enterprises

Judges

- Faculty Members
 - Industry Experts
 - Mentors
-

6. User Roles

Participant

- Register
- Login
- Create Team
- Join Team

- Submit Project
 - Upload Repository
 - Upload PPT
 - Upload Demo Video
 - View Scores
 - View Leaderboard
-

Organizer

- Create Hackathon
 - Manage Participants
 - Manage Teams
 - Configure Evaluation Rubric
 - Monitor Submissions
 - View Analytics
 - Publish Results
-

Judge

- View AI Reports
 - Review Projects
 - Override Scores (Optional)
 - Add Feedback
 - Finalize Results
-

7. Functional Requirements

Module 1 – Authentication

Features

- User Registration
 - Login
 - JWT Authentication
 - Password Encryption
 - Forgot Password
 - Email Verification
-

Module 2 – Participant Dashboard

Features

- Join Hackathon
 - Team Management
 - Submission Status
 - Notifications
 - Profile Management
-

Module 3 – Team Management

Features

- Create Team
 - Invite Members
 - Accept Invitations
 - Team Dashboard
 - Team Progress
-

Module 4 – Project Submission

Participants can upload:

- GitHub Repository Link
 - ZIP File
 - PPT
 - PDF Documentation
 - Demo Video
 - README
 - Tech Stack
 - Live Deployment URL
-

Module 5 – AI Project Evaluation

The AI analyzes:

- Source Code
- README
- Project Description
- Documentation
- PPT

- Demo Video

Evaluation Parameters

- Innovation
- Technical Complexity
- Feature Completeness
- UI/UX
- Business Impact
- Scalability
- Documentation
- Code Quality
- Architecture
- Presentation Quality

Output

- Overall Score
- Individual Scores
- AI Feedback
- Improvement Suggestions

Module 6 – Plagiarism Detection

The system compares:

- Folder Structure
- Source Files
- Function Similarity
- README Similarity
- Class Structure
- Project Architecture

Outputs

- Similarity Percentage
- Duplicate Files
- Matching Code Sections
- Risk Level

Module 7 – AI Generated Code Detection

The engine estimates whether a project relies heavily on AI-generated code by analyzing:

- Coding Style Consistency

- Repetitive Patterns
- Project Structure
- Documentation Consistency

Output

- Estimated AI Usage Percentage
 - Confidence Score
 - Risk Assessment
-

Module 8 – Static Code Analysis

Checks include:

- Bugs
- Security Issues
- Complexity
- Code Smells
- Dead Code
- Maintainability

Reports Generated

- Static Analysis Report
 - Security Report
 - Code Quality Score
-

Module 9 – Secure Docker Sandbox

Each project executes inside an isolated Docker container.

Workflow

1. Clone Repository
2. Build Project
3. Install Dependencies
4. Execute Application
5. Run Test Cases
6. Capture Logs
7. Measure Performance
8. Destroy Container

Metrics

- Build Status

- Execution Time
 - CPU Usage
 - Memory Usage
 - Runtime Errors
-

Module 10 – PPT Analysis

AI extracts:

- Problem Statement
- Solution
- Architecture
- Innovation
- Market Potential
- Business Model

Outputs

- Presentation Score
 - Missing Sections
 - Improvement Suggestions
-

Module 11 – Demo Video Analysis

The system performs:

- Speech-to-Text
- Transcript Generation
- Feature Extraction
- Presentation Analysis

Outputs

- Transcript
 - Communication Score
 - Feature Coverage
 - Confidence Score
-

Module 12 – Leaderboard

Displays

- Team Rank

- Overall Score
- Technical Score
- Innovation Score
- UI Score
- Impact Score

Updates automatically after evaluation.

Module 13 – Organizer Dashboard

Features

- Total Participants
 - Total Teams
 - Submission Statistics
 - Fraud Alerts
 - Evaluation Progress
 - Leaderboard
 - Reports
 - Export Results
-

Module 14 – Judge Dashboard

Features

- AI Evaluation Report
 - Manual Review
 - Score Override
 - Comments
 - Final Decision
-

8. Evaluation Rubric

Parameter	Weight
Technical Complexity	30%
Innovation	20%
UI/UX	15%
Business Impact	15%
Documentation	10%

Parameter	Weight
Presentation	10%

9. Non-Functional Requirements

- Responsive UI
 - Secure Authentication
 - Role-Based Access Control
 - Fast API Response
 - Container Isolation
 - Data Encryption
 - Audit Logging
 - High Availability
 - Scalable Architecture
 - Modular Backend
-

10. Technology Stack

Frontend

- React
 - TypeScript
 - Vite
 - Tailwind CSS
 - Shadcn UI
 - Framer Motion
-

Backend

- FastAPI
 - Python
-

Database

- PostgreSQL
-

Authentication

- JWT
-

AI Services

- Gemini API (Free Tier)
 - Whisper (Speech-to-Text)
-

Static Analysis

- Semgrep
 - ESLint
 - Pylint
-

Sandbox

- Docker
-

Storage

- Local Storage
 - MinIO (Optional)
 - AWS S3 (Optional)
-

Deployment

- Vercel (Frontend)
 - Render (Backend)
 - Docker
-

11. System Workflow

1. Organizer creates hackathon.
2. Participants register and create teams.
3. Teams submit repository and project assets.
4. Platform performs plagiarism detection.

5. Static code analysis begins.
 6. Project executes inside Docker sandbox.
 7. AI evaluates repository, documentation, PPT, and demo video.
 8. Individual scores are generated.
 9. Final weighted score is calculated.
 10. Leaderboard updates automatically.
 11. Judges review AI reports (optional).
 12. Winners are announced.
-

12. Success Metrics

- 90% reduction in evaluation time.
 - Less than 5 minutes average evaluation per submission.
 - Automated scoring for all projects.
 - Real-time leaderboard updates.
 - Detection of suspicious or duplicate submissions.
 - Improved consistency and transparency in judging.
-

13. Future Enhancements

- Multi-language code support.
 - Custom evaluation rubrics.
 - AI mentor recommendations.
 - Live coding challenge integration.
 - Certificate generation.
 - Sponsor dashboard.
 - Blockchain-based submission verification.
 - Advanced Git commit history analysis.
 - Multi-cloud sandbox execution.
 - Public API for hackathon integrations.
-

14. Assumptions & Limitations

Assumptions

- Projects are submitted through GitHub or ZIP archives.
- Docker is available on the execution server.
- AI APIs operate within free-tier limits during development.

Limitations

- Public GitHub-wide plagiarism detection is outside the scope of the MVP.
 - AI-generated code detection provides an estimate rather than definitive proof.
 - Large-scale concurrent sandbox execution requires additional infrastructure.
-

15. Minimum Viable Product (MVP)

The initial release will include:

- User Authentication
 - Team Management
 - Hackathon Creation
 - Project Submission
 - AI Repository Evaluation
 - README Analysis
 - Static Code Analysis
 - Docker Sandbox Execution
 - Basic Plagiarism Detection (between submissions)
 - AI Feedback Report
 - Organizer Dashboard
 - Live Leaderboard
-

16. Expected Outcome

HackGuard AI will provide a secure, AI-assisted, and scalable platform that streamlines hackathon operations by automating project evaluation, improving transparency, reducing manual effort, and helping organizers deliver fair and efficient results.